

National Forensic Sciences University
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National Forensic Sciences University, Gandhinagar

School of Cyber Security and Digital Forensics

Assignment 4: Docker & Container

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Sem: 3
Subject: Cloud Security Forensics
Course: MSc Cyber Security
Batch: 2025 -2027

Install Docker Engine & Create Docker Hub Account

Docker Engine is the core background service (daemon) that builds and runs containers; the Docker CLI is the command-line tool used to talk to it.

Docker Hub is Docker's public online registry for storing and sharing container images, similar to how GitHub hosts code repositories.





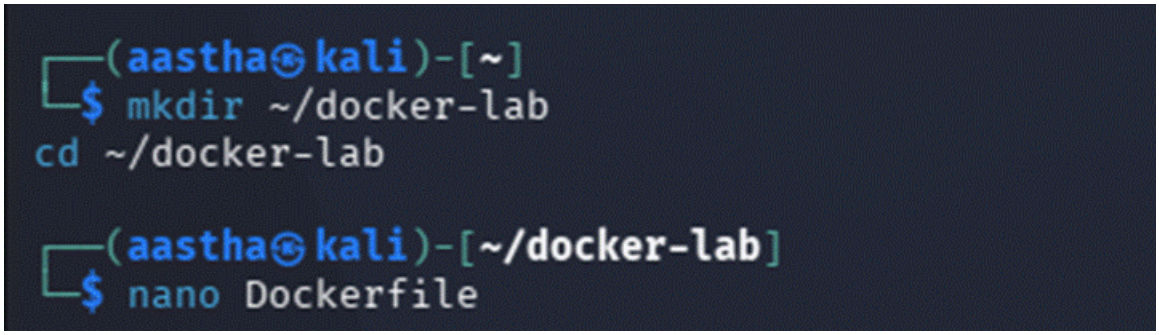
1. Pull the Docker Image 'alpine' from Docker Hub

Alpine is a very small, security-focused Linux distribution (its base image is only a few MB), commonly used as a lightweight base for containers. 'docker pull <image>' downloads an image from a registry (Docker Hub by default) to the local machine. 'docker images' lists images already stored locally, and 'docker run <image>' starts a new container from an image.



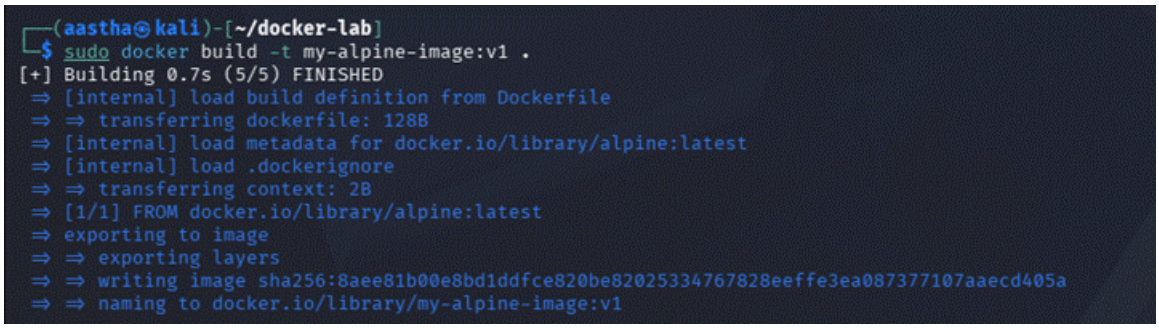
2. Build an Image from a Dockerfile

A Dockerfile is a text file containing step-by-step instructions for assembling a custom image. 'FROM' sets the base image, 'LABEL' attaches metadata (such as a maintainer name) to the image, and 'CMD' specifies the default command a container runs when started.




```
FROM alpine:latest
 LABEL maintainer="Aastha"
 CMD ["echo", "Hello from my Docker image!"]
```

'docker build -t <name>:<tag> .' reads the Dockerfile in the current directory ('.') and builds a new image from it, tagging it with the given name and version.



'docker images' again confirms the newly built image now exists locally, and 'docker run --rm <image>' runs a container from it and automatically deletes the container once it exits.

```
┌──(aastha㉿kali)-[~/docker-lab]
└─$ sudo docker images
REPOSITORY        TAG       IMAGE ID       CREATED         SIZE
alpine            latest    d529dd0c6e55   2 months ago    8.42MB
my-alpine-image   v1        8aee81b00e8b   2 months ago    8.42MB

┌──(aastha㉿kali)-[~/docker-lab]
└─$ sudo docker run --rm my-alpine-image:v1
Hello from my Docker image!
```

3. Create an Image with My Name and Push It to the Repository

'docker login' authenticates the CLI with a Docker Hub account so images can be pushed to it.

```
┌──(aastha㉿kali)-[~/docker-lab]
└─$ sudo docker login

USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login -u <username>'


Your one-time device confirmation code is: KKRD-FFTV
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser…

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
```

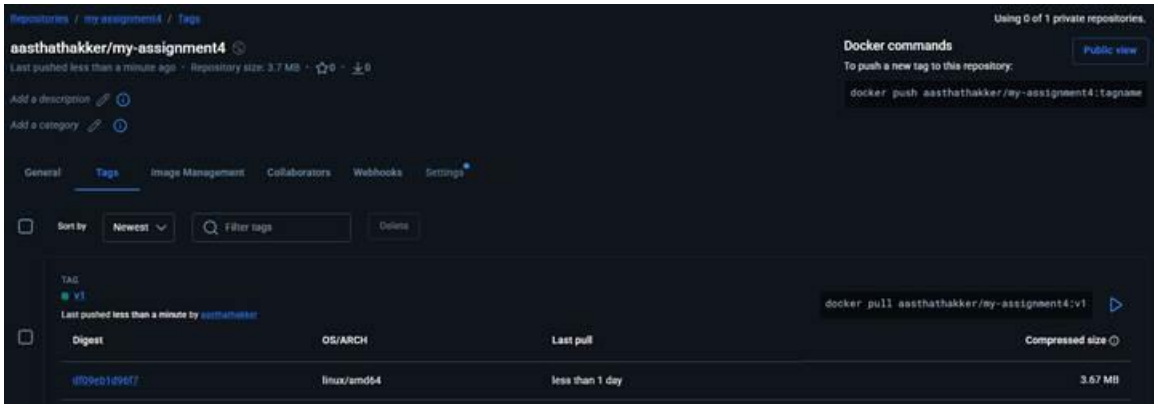
Before pushing, an image must be tagged in the form '<dockerhub-username>/<repository>:<tag>', which is how Docker Hub identifies which account and repository an image belongs to.

```
┌──(aastha㉿kali)-[~/docker-lab]
└─$ sudo docker images
REPOSITORY                      TAG       IMAGE ID       CREATED          SIZE
<none>                          <none>    ce987fee5186   7 minutes ago    1.12GB
aastha-python                   v1        b5f0188be611   7 minutes ago    1.12GB
python                          latest    691aea61c5aa   4 hours ago      1.12GB
alpine                          latest    d529dd0c6e55   2 months ago     8.42MB
aasthathakker/my-assignment4    v1        8aee81b00e8b   2 months ago     8.42MB
aasthathakker/my-docker-image   v1        8aee81b00e8b   2 months ago     8.42MB
my-alpine-image                 v1        8aee81b00e8b   2 months ago     8.42MB
```

'docker push <username>/<repository>:<tag>' uploads the tagged image to Docker Hub, making it available for others to pull.

```
┌──(aastha㉿kali)-[~/docker-lab]
└─$ sudo docker push aasthathakker/my-assignment4:v1
The push refers to repository [docker.io/aasthathakker/my-assignment4]
34884abbe928: Mounted from library/alpine
v1: digest: sha256:df09eb1d96f7dc5623e0d90e5c4f91249b0838342a5e638864bcbaa9e84b3841 size: 527
```

Docker Hub's repository page shows the pushed image, along with its tag, digest (a unique content hash identifying that exact image), architecture, and size.



4. List All Docker Containers with Container Details

A container is a running (or stopped) instance of an image. 'docker ps' by itself only shows currently running containers; adding '-a' shows all containers regardless of state, along with their ID, image, command, creation time, status, ports, and name.

```
┌──(aastha㉿kali)-[~/docker-lab]
└─$ sudo docker ps -a
CONTAINER ID   IMAGE     COMMAND     CREATED             STATUS                          PORTS     NAMES
6a267ba587a4   python    "python3"   13 minutes ago      Exited (0) 11 minutes ago                 epic_yalow
f85fc607de9e   python    "bash"      13 minutes ago      Exited (0) 13 minutes ago                 relaxed_elbakyan
b538fe4ca4cf   alpine    "/bin/sh"   About an hour ago   Exited (130) 58 minutes ago               sweet_sammet
b0cbf53787bb   alpine    "/bin/sh"   About an hour ago   Exited (0) About an hour ago              kind_jang
```

5. Detailed Information about the Docker Installation

'docker info' reports system-wide details about the Docker installation itself — such as how many containers and images exist, the storage driver in use, and the cgroup driver/version used to manage resources.

```
┌──(aastha㉿kali)-[~/docker-lab]
└─$ sudo docker info
Client:
 Version:    28.5.2+dfsg4
 Context:    default
 Debug Mode: false
 Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  0.29.1+ds1
    Path:     /usr/libexec/docker/cli-plugins/docker-buildx

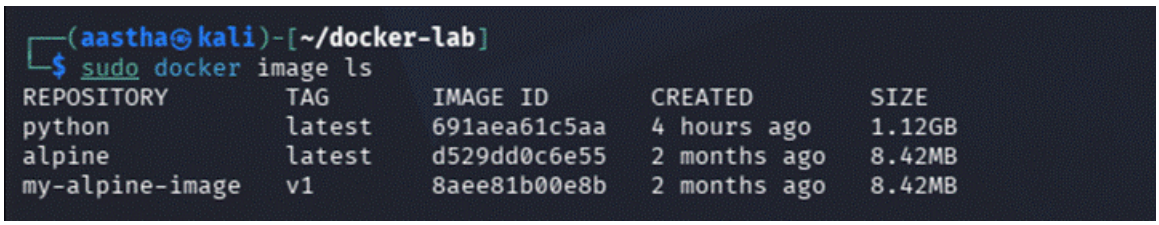
Server:
 Containers: 2
  Running: 0
  Paused: 0
  Stopped: 2
 Images: 3
 Server Version: 28.5.2+dfsg4
 Storage Driver: overlay2
  Backing Filesystem: extfs
  Supports d_type: true
  Using metacopy: false
  Native Overlay Diff: true
```

'docker version' shows version details for both the Client (CLI) and the Server (daemon/Engine), including the API version, the Go language version it was built with, and the build date.



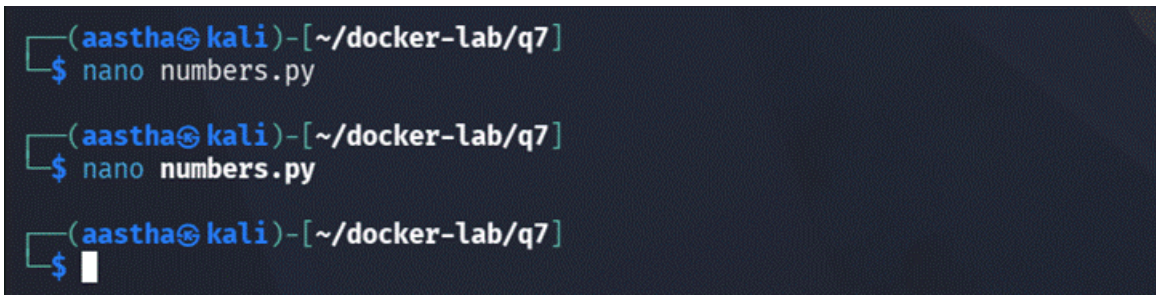
6. List All Docker Images with Details (TAG / IMAGE ID / SIZE)

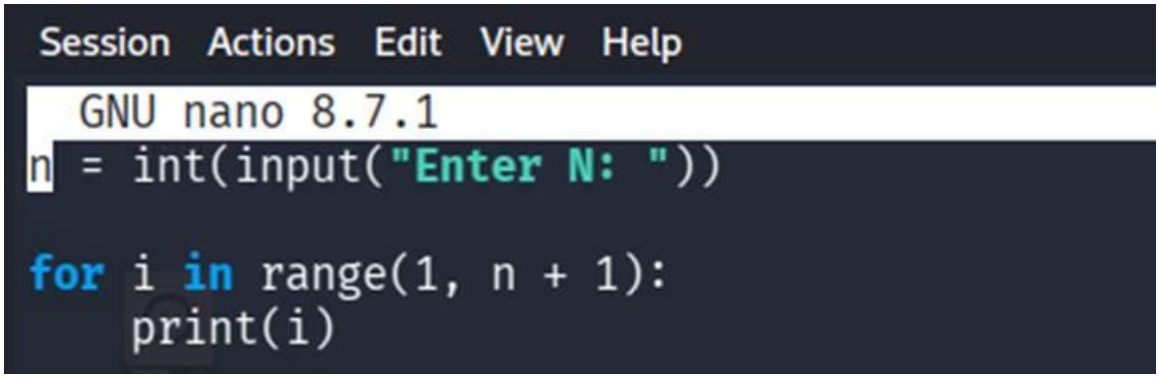
An image is a read-only template used to create containers. 'docker image ls' (equivalent to 'docker images') lists every image stored locally, showing its repository name, tag (a label such as 'latest' or 'v1' identifying a version), unique Image ID, creation date, and disk size.



7. Simple Python Program to Print N Numbers

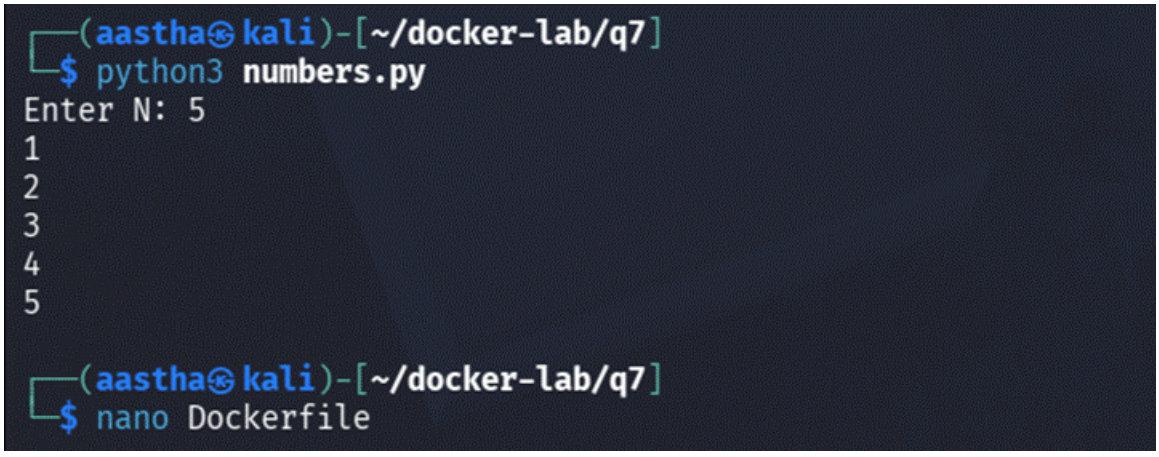
nano is a simple command-line text editor used here to create a Python script directly from the terminal.





Test it by running, and create a Dockerfile

'python3 <file>.py' runs the script directly on the host machine, without Docker, to confirm the code itself works. A Dockerfile was then written so the same script can be packaged and run inside a container.





In this Dockerfile, 'WORKDIR /app' sets the working directory inside the image, 'COPY' copies the script into it, and 'CMD' defines the command that runs the script when a container starts.

```
┌──(aastha㉿kali)-[~/docker-lab/q7]
└─$ sudo docker build -t python-numbers:v1 .
[sudo] password for aastha:

[+] Building 8.9s (9/9) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 122B
 => [internal] load metadata for docker.io/library/python:3.12-slim
 => [auth] library/python:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/3] FROM docker.io/library/python:3.12-slim@sha256:7a8b475003c4fe15a2cd4e55e5cfc2f3560bdc9333d624f24cdd6d4340fd7a17
 => => resolve docker.io/library/python:3.12-slim@sha256:7a8b475003c4fe15a2cd4e55e5cfc2f3560bdc9333d624f24cdd6d4340fd7a17
 => => sha256:b79f58b3eb945fe861ace4fc38504483c5b100c794aa095b6c9f112ed9807aad 12.12MB / 12.12MB
 => => sha256:7a8b475003c4fe15a2cd4e55e5cfc2f3560bdc9333d624f24cdd6d4340fd7a17 10.37kB / 10.37kB
 => => sha256:a249c9f47e05708dd367f3fe8ada03cf347390fad66fb8b0518c0ef55ae3cb84 1.75kB / 1.75kB
 => => sha256:72a58063c7563c5c64052da2becebce3f25b60055cb351b668c4e2df9153702c 5.65kB / 5.65kB
 => => sha256:6310eb16bf4251731feab01e8f633bf5e2d75a657ccad97f420b1f83cce457be 29.79MB / 29.79MB
 => => sha256:a0d8783128af79059fe015e6313ce9e8fd7856b194e57f2d3c909b12a593bd2f 1.29MB / 1.29MB
 => => sha256:48579d87405c21ba2599e244fcf3a02e900dbf9b9103e4afd030fbe0c7f43b33 250B / 250B
 => => extracting sha256:6310eb16bf4251731feab01e8f633bf5e2d75a657ccad97f420b1f83cce457be
 => => extracting sha256:a0d8783128af79059fe015e6313ce9e8fd7856b194e57f2d3c909b12a593bd2f
 => => extracting sha256:b79f58b3eb945fe861ace4fc38504483c5b100c794aa095b6c9f112ed9807aad
 => => extracting sha256:48579d87405c21ba2599e244fcf3a02e900dbf9b9103e4afd030fbe0c7f43b33
 => [internal] load build context
 => => transferring context: 105B
 => [2/3] WORKDIR /app
 => [3/3] COPY numbers.py .
 => exporting to image
 => => exporting layers
 => => writing image sha256:065ea7b643ed8e64fcba23e06dd9ca0b85c13eb97f1a8d5e59f2157ad0abf024
 => => naming to docker.io/library/python-numbers:v1
```

'docker run -it --rm <image>' runs the container interactively ('-it' keeps the terminal attached so input() works), removing the container automatically afterwards ('--rm').

```
┌──(aastha㉿kali)-[~/docker-lab/q7]
└─$ sudo docker run -it --rm python-numbers:v1
Enter N: 5
1
2
3
4
5
```

8. Search for a Docker Image on Docker Hub with My Name

'docker search <term>' queries Docker Hub for public repositories matching that term, without downloading anything — it simply lists matching image names, descriptions, star counts, and whether each is an official image.

```
┌──(aastha㉿kali)-[~/docker-lab/q7]
└─$ sudo docker search aasthathakker
NAME                            DESCRIPTION   STARS     OFFICIAL
aasthathakker/my-assignment4                  0
```

9. Create and Run a Docker Container for a Nginx Web Server

Pull nginx

Nginx is a widely used, lightweight web server; its official Docker image serves a default welcome page out of the box.

```
┌──(aastha㉿kali)-[~/docker-lab/q8]
└─$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
6310eb16bf42: Already exists
30576ad53d33: Pull complete
b8f66660faa6: Pull complete
657dd7fba849: Pull complete
c90544874aaf: Pull complete
8f655e1bd5c1: Pull complete
0a35a4e59186: Pull complete
Digest: sha256:b34848eff6db786b6b1282d3a9c3fd0b5563dfb6d261df4923378b419e0d24f0
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

Create a container

'docker run -d --name <name> -p 8080:80 <image>' starts a container in the background ('-d' for detached mode), names it, and maps port 8080 on the host to port 80 inside the container (where nginx listens by default).

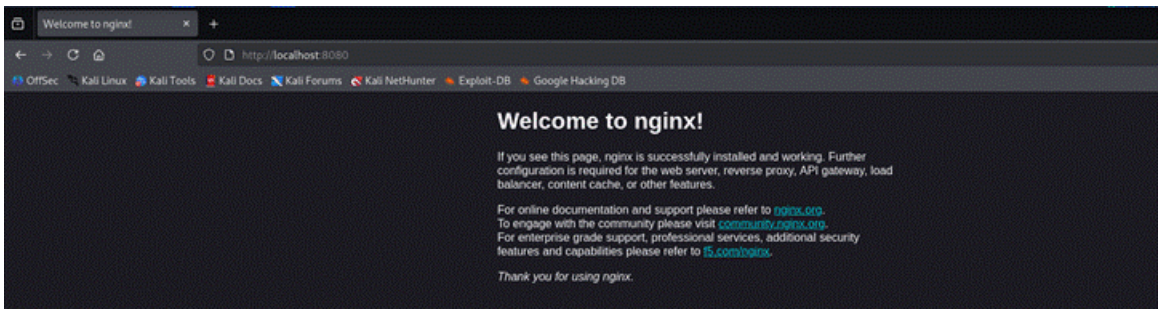
```
┌──(aastha㉿kali)-[~/docker-lab/q8]
└─$ sudo docker run -d --name my-nginx -p 8080:80 nginx
1283d2a37ee5489f82f900596e78c72150ff003fd9c1177e7a0b693e1f593b40
```

Check container

```
┌──(aastha㉿kali)-[~/docker-lab/q8]
└─$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                     NAMES
1283d2a37ee5   nginx     "/docker-entrypoint.…"   47 seconds ago   Up 47 seconds   0.0.0.0:8080→80/tcp, [::]:8080→80/tcp   my-nginx
```

Open URL in browser

Because port 8080 on the host is mapped to port 80 in the container, the running web server can be reached at http://localhost:8080 in a browser.



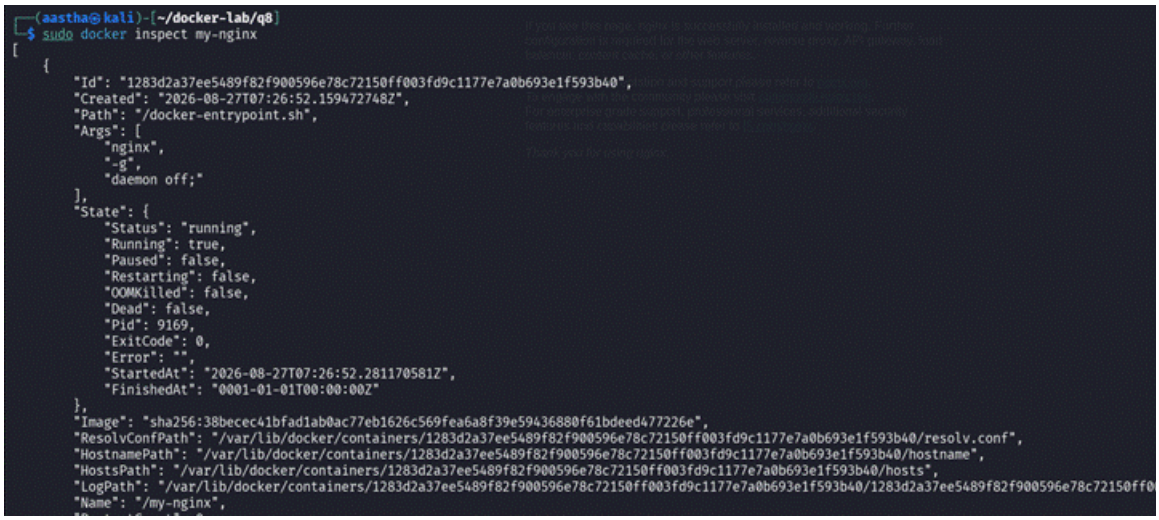
Check logs

'docker logs <container>' prints the output/log stream produced by the process running inside a container. Here, nginx's startup messages and access log entries.

```
┌──(aastha㉿kali)-[~/docker-lab/q8]
└─$ sudo docker logs my-nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/08/27 07:26:52 [notice] 1#1: using the "epoll" event method
2026/08/27 07:26:52 [notice] 1#1: nginx/1.31.4
2026/08/27 07:26:52 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/08/27 07:26:52 [notice] 1#1: OS: Linux 6.18.12+kali-amd64
2026/08/27 07:26:52 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
2026/08/27 07:26:52 [notice] 1#1: start worker processes
2026/08/27 07:26:52 [notice] 1#1: start worker process 29
2026/08/27 07:26:52 [notice] 1#1: start worker process 30
172.17.0.1 - - [27/Aug/2026:07:28:22 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0" "-"
```

Inspect

'docker inspect <container>' returns detailed low-level configuration and state information about a container, formatted as JSON — including its ID, status, process ID, and internal file paths.



'docker start' and 'docker stop' pause/resume a container without deleting it, while 'docker rm -f' forcibly removes a container (stopping it first if needed) so it no longer exists.

```
┌──(aastha㉿kali)-[~/docker-lab/q8]
└─$ sudo docker start my-nginx
my-nginx

┌──(aastha㉿kali)-[~/docker-lab/q8]
└─$ sudo docker stop my-nginx
my-nginx
```

```
┌──(aastha㉿kali)-[~/docker-lab/q8]
└─$ sudo docker rm -f my-nginx
my-nginx
```

10. Automated Pentesting Container (OWASP ZAP) and Report Generation

OWASP ZAP (Zed Attack Proxy) is an open-source web application security scanner used to automatically find vulnerabilities such as missing security headers, information disclosure, and injection risks. It is referenced from https://github.com/zaproxy/zaproxy and https://github.com/vishnudxb/automated-pentest.

Pull the ZAP Docker image

```
┌──(aastha㉿kali)-[~/docker-lab]
└─$ sudo docker pull zaproxy/zap-stable
Using default tag: latest
latest: Pulling from zaproxy/zap-stable
039e6f9f9752: Pull complete
d6b9d96ab582: Pull complete
a3cc44adff2f: Pull complete
b4d37c583758: Pull complete
bd527f0cd38b: Pull complete
d216f30309c1: Pull complete
4f4fb700ef54: Pull complete
0d2cbb6a3c69: Pull complete
c0b230616ec7: Pull complete
7e90f09586cd: Pull complete
f508357c279f: Pull complete
16465afc17cf: Pull complete
```

Create a container from the ZAP image

'docker create' sets up a container from an image without starting it yet — this differs from 'docker run', which creates and starts a container in one step.

```
┌──(aastha㉿kali)-[~/docker-lab]
└─$ sudo docker create --name zap-pentest zaproxy/zap-stable
3e66e6f288495b38294b47bb764469ae0a41a5a2ae4575babecfc04e35083852

┌──(aastha㉿kali)-[~/docker-lab]
└─$ sudo docker ps -a
CONTAINER ID   IMAGE                  COMMAND                  CREATED          STATUS                       PORTS                                      NAMES
3e66e6f28849   zaproxy/zap-stable     "bash"                   35 seconds ago   Created                                                         zap-pentest
76b63a8f209e   aastha-alpine-web:v1   "/docker-entrypoint.…"   2 days ago       Exited (255) 29 minutes ago  0.0.0.0:8080→80/tcp, [::]:8080→80/tcp  aastha-web
6a267ba587a4   python                 "python3"                2 days ago       Exited (0) 2 days ago                                           epic_yalow
f85fc607de9e   python                 "bash"                   2 days ago       Exited (0) 2 days ago                                           relaxed_elbakyan
b538fe4ca4cf   alpine                 "/bin/sh"                2 days ago       Exited (130) 2 days ago                                         sweet_sammet
b0cbf53787bb   alpine                 "/bin/sh"                2 days ago       Exited (0) 2 days ago                                           kind_jang
```

Start it

```
┌──(aastha㉿kali)-[~/docker-lab]
└─$ sudo docker start zap-pentest
zap-pentest
```

Create a directory for the pentesting report

A local folder is created to hold the scan report files that ZAP will generate.

```
┌──(aastha㉿kali)-[~/docker-lab]
└─$ mkdir -p ~/zap-pentest
cd ~/zap-pentest
```

The '-v "$(pwd):/zap/wrk/:rw"' option mounts (shares) the local folder into the container so files written inside it also appear on the host. 'zap-baseline.py' is ZAP's automated quick-scan script, run here against a test target, with '-r/-w/-x/-J' each requesting the report in a different format (HTML, Markdown, XML, JSON).

```
┌──(aastha㉿kali)-[~/zap-pentest]
└─$ sudo docker run --rm \
  -v "$(pwd):/zap/wrk/:rw" \
  -t zaproxy/zap-stable \
  zap-baseline.py \
  -t http://testasp.vulnweb.com \
  -r zap-report.html \
  -w zap-report.md \
  -x zap-report.xml \
  -J zap-report.json
Using the Automation Framework
Total of 20 URLs
PASS: Vulnerable JS Library (Powered by Retire.js) [10003]
PASS: In Page Banner Information Leak [10009]
PASS: Cookie Without Secure Flag [10011]
PASS: Re-examine Cache-control Directives [10015]
PASS: Cross-Domain JavaScript Source File Inclusion [10017]
PASS: Content-Type Header Missing [10019]
PASS: Information Disclosure - Sensitive Information in URL [10024]
```

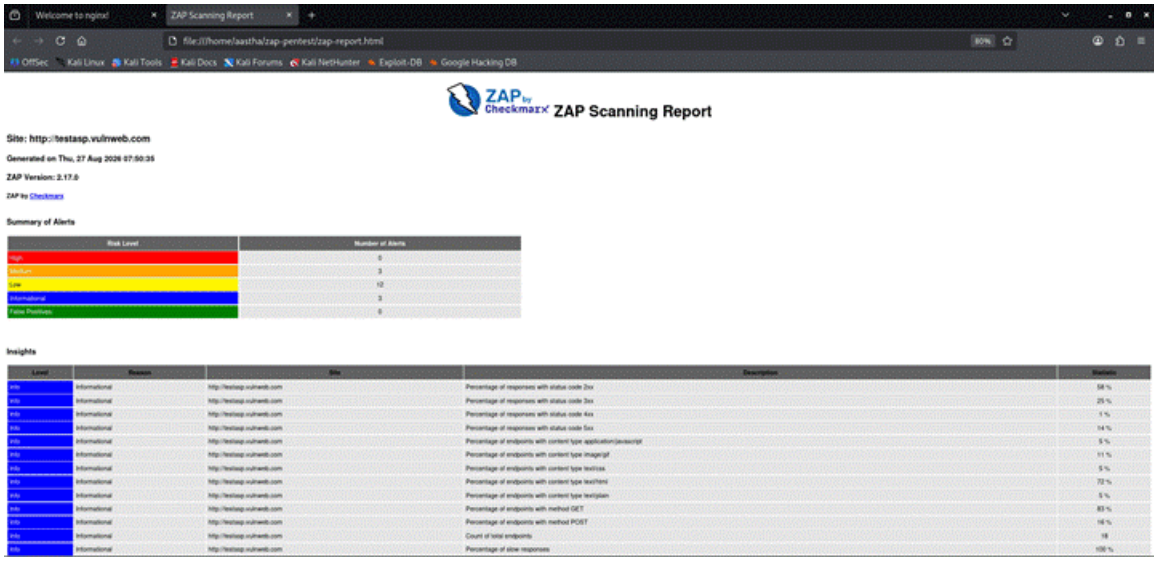
Open HTML report

'xdg-open' launches the default application (a browser, for an .html file) to view a file.

```
┌──(aastha㉿kali)-[~/zap-pentest]
└─$ ls
zap-report.html  zap-report.json  zap-report.md  zap-report.xml  zap.yaml

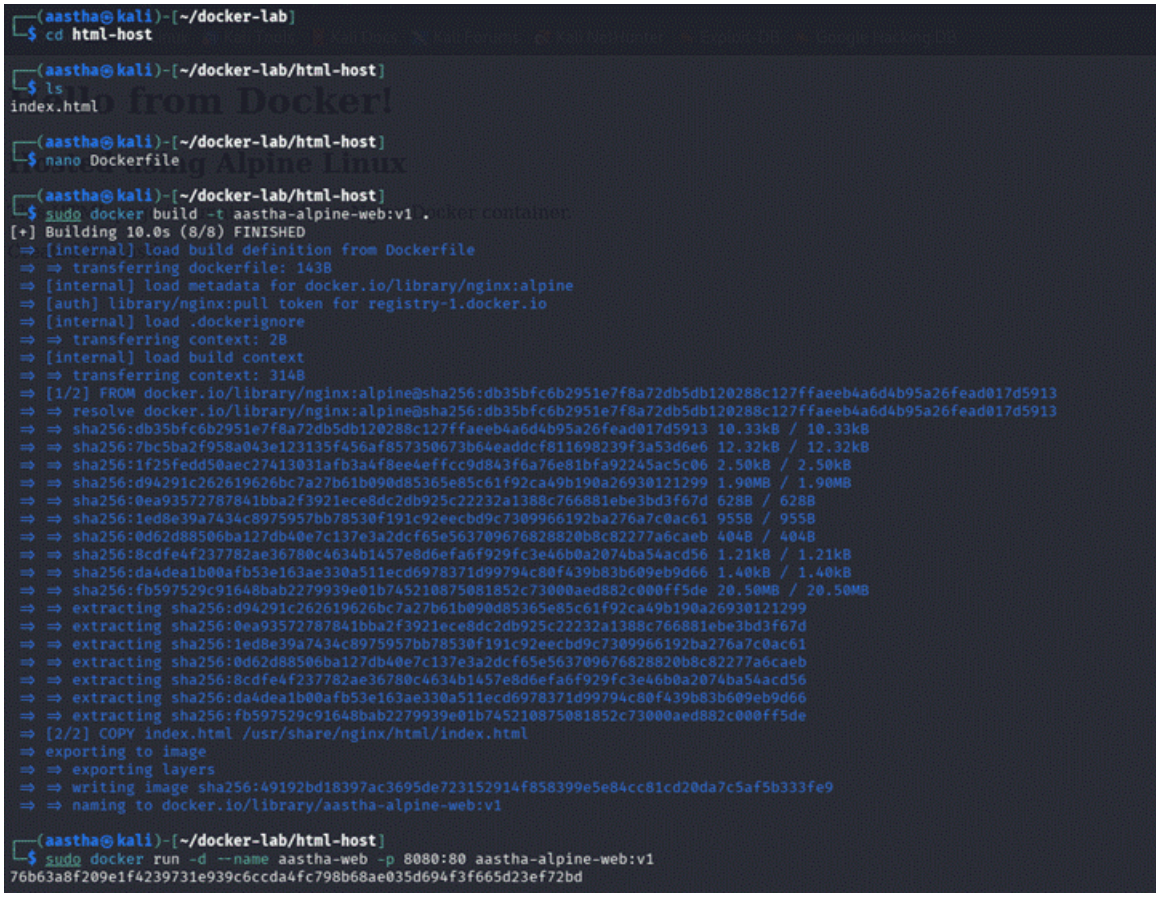
┌──(aastha㉿kali)-[~/zap-pentest]
└─$ xdg-open zap-report.html
```

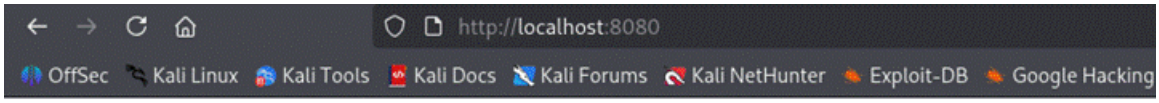
The report summarizes findings by risk level — High, Medium, Low, and Informational — along with details for each alert found on the scanned site.



Additional demonstration — custom HTML page served via a Docker container

A custom static HTML page was packaged into a lightweight nginx:alpine-based image and run as its own container, reachable the same way as the earlier nginx example.





Hello from Docker!

Hosted using Alpine Linux

This HTML page is running inside an Nginx Docker container.

Created by Aastha